

TRANSVEXA · TEAM LEADER KEYNOTE & 20-MINUTE SIH PITCH SCRIPT

Problem Statement: SIH26154 | **Organization:** NTRO | **Theme:** Blockchain & Cybersecurity | **Team:** 6 Members

1. Team Leader Opening Keynote Speech (2 Minutes)

Team Leader: 'Respected Members of the Jury, NCIIPC and NTRO Representatives, and distinguished evaluators-a very warm good morning.'

Team Leader: 'My name is [Leader Name], and I am proud to lead Team Transvexa. Today, we stand before you to solve Problem Statement SIH26154: Gen AI Platform for Automated Content Transformation.'

Team Leader: 'In national security operations, cyber threats move at machine speed. When a zero-day vulnerability hits critical infrastructure, our defense agencies cannot afford a 4-hour delay spent manually translating, reformatting, and adapting threat advisories for different agencies, CISOs, and the public.'

Team Leader: 'We built Transvexa to solve this exact problem: A secure, enterprise-grade AI transformation platform that converts a single raw incident report into 8 standardized, cryptographically verified operational deliverables in under 10 seconds.'

Team Leader: 'Over the next 18 minutes, our specialized 6-member team will walk you through our platform: My Frontend leads will show you the live terminal workspace; my Backend leads will explain our FastAPI architecture and Blockchain verification ledger; and my AI/ML leads will detail our OCR pipeline, PII sanitization, and local air-gapped LLM deployment.'

Team Leader: 'Let us begin with our Frontend & UI Walkthrough. Member 2, over to you!'

2. Master Presentation Agenda (20 Minutes Total)

- 0:00 - 2:00 (2 Min): Team Leader Opening Keynote (Team Leader)
- 2:00 - 6:00 (4 Min): Phase 1 & 2 · Live Transvexa SOC Terminal & 8 Deliverables Demo (Member 2)
- 6:00 - 10:00 (4 Min): Phase 3 · Backend Architecture & Database Design (Member 3)
- 10:00 - 14:00 (4 Min): Phase 4 · Cybersecurity & Blockchain Verification Engine (Member 4)
- 14:00 - 17:00 (3 Min): Phase 5 · AI/ML, NLP & Local Air-Gapped LLM Architecture (Members 5 & 6)
- 17:00 - 20:00 (3 Min): Phase 6 · Team Leader Closing Pitch & Open Q&A (Team Leader & Team)

3. Domain Presentations

PHASE 2: LIVE TRANSVEXA PLATFORM WALKTHROUGH (2:00 - 6:00) · Member 2

[Action: Member 2 loads Transvexa SOC Terminal, uploads sample PDF incident report, and shows extracted text]

Member 2: 'Thank you, Leader. On our SOC Terminal workspace, an operator submits source content by uploading a PDF incident log or pasting raw text into the left panel.'

Member 2: 'On the right panel, our Configurable Transformation Matrix allows selecting 11 Indian regional languages-such as Hindi, Tamil, or Bengali-and audience levels: System Level (SOC), Organization Level (CISO), or People Level (Citizen Safety).'

[Action: Member 2 clicks 'Select All' for output formats and hits 'Run Analysis']

Member 2: 'Transvexa performs simultaneous batch transformation into all 8 required deliverable formats: CERT-In Advisory, Executive Summary, Action Plan, Video Package Storyboard, Twitter Thread, Infographic Blueprint, Presentation Slide Deck with speaker notes, and LinkedIn Briefing. Each card supports instant 1-click Copy and Export.'

PHASE 3: BACKEND ARCHITECTURE & DATABASE DESIGN (6:00 - 10:00) · Member 3

Member 3: 'Behind this terminal is an asynchronous FastAPI backend deployed on Python. When an operator triggers analysis, our API server manages concurrent batch execution without blocking looped threads.'

Member 3: 'Our database layer utilizes SQLAlchemy ORM backed by PostgreSQL on Supabase, maintaining relational persistence across User, Upload, and AnalysisResult models with JWT token security.'

PHASE 4: CYBERSECURITY & BLOCKCHAIN VERIFICATION ENGINE (10:00 - 14:00) · Member 4

[Action: Member 4 points to Export / Blockchain Verification badge on screen]

Member 4: 'To fulfill the Blockchain & Cybersecurity theme, we eliminate Data Tampering: When an advisory is finalized, our backend computes a deterministic SHA-256 hash of the content and anchors it to an immutable Smart Contract Blockchain ledger (Polygon / Hyperledger). Any agency can verify advisory authenticity against the ledger in real time.'

PHASE 5: AI/ML, NLP & LOCAL LLM ARCHITECTURE (14:00 - 17:00) · Members 5 & 6

Member 5: 'We built a dual-stage ingestion engine using PyMuPDF for digital text extraction with automatic Tesseract OCR fallback for scanned images, preceded by an automated PII & Sensitive Data Sanitization Filter that redacts internal IPs and credentials.'

Member 6: 'We enforce strict JSON Schema prompts to eliminate hallucinations, and our decoupled backend allows instant zero-code swapping to on-premise quantized local LLMs (Llama-3 via Ollama) inside air-gapped NTRO facilities.'

4. Team Leader Closing Remarks (17:00 - 20:00)

Team Leader: 'Thank you team. Respected Judges, to summarize what we have presented today:'

1. Complete PS 26154 Compliance: Simultaneous 1-click batch generation of all 8 deliverable formats.
2. Operational Impact: 98% reduction in content creation time (from 4 hours to 10 seconds).
3. Blockchain Integrity: Non-repudiation and SHA-256 tamper-proof ledger anchoring.
4. Defense Readiness: Air-gapped local LLM compatibility and PII sanitization for NTRO facilities.

Team Leader: 'Transvexa is not just a hackathon prototype-it is a production-architected, security-first platform ready for national defense deployment. Thank you for your time, and we are now honored to take your questions!'

5. Extended Jury Q&A Response Matrix

Q: Why generate storyboards/scripts for video instead of full AI video rendering?

Team Leader & Member 2: 'Emergency threat advisories require rapid 10-second turnarounds. Full AI video rendering takes several minutes and produces massive video files. A structured Video Package-with scene shots, narration text, subtitles, and motion graphic prompts-allows production and broadcast teams to immediately publish across TV channels and social media without delay.'

Q: How does the system handle high concurrent user load during major cyber incidents?

Member 3: 'FastAPI's asynchronous ASGI architecture handles non-blocking I/O operations efficiently. Our /analyze endpoint processes deliverable tasks in parallel worker threads, and PostgreSQL connection pooling (pool_pre_ping=True) ensures database stability under peak load.'

Q: Where and how is the Blockchain implemented?

Member 4: 'When an advisory is generated, we calculate a SHA-256 hash of the JSON content. This hash, along with timestamp and operator ID, is signed and published to a smart contract registry (e.g. Polygon testnet / Hyperledger). If anyone alters even a single character in the DB, the hash changes, failing verification.'

Q: How do you prevent AI hallucinations when generating technical CVE IDs or threat metrics?

Member 5: 'We enforce strict JSON extraction schemas that restrict the model to extracting explicitly mentioned CVEs and IOCs from the source text. For unmentioned fields, post-processing validation flags them, and our roadmap integrates a deterministic cross-check against National Vulnerability Databases (NVD).'

Q: How do you handle sensitive defense data sent to cloud AI APIs?

Member 6 & Team Leader: 'First, our regex sanitizer automatically masks internal IPs, credentials, and phone numbers before API dispatch. Second, the backend AI layer is decoupled via a standardized interface, allowing zero-code swapping between cloud APIs and local, air-gapped LLMs like Llama-3 running on internal GPU servers.'